

ENVR 405: Water Resources Engineering and Hydrology

Term: Fall 2026 | Instructor: Dr. Luna Leopold (lleopold@envsci.univ.edu)

Course Description & Overview:

Hydrological cycle modeling, open-channel flood routing, groundwater hydraulics, and watershed management under climate stress.

COURSE ASSIGNMENTS & WEIGHT DISTRIBUTION

Assignment Description	Weight	Due Module	Deliverable Format
Watershed Delineation & Runoff Hydrograph Lab	25%	Week 04	GIS Model & Calculations
HEC-HMS Flood Inundation Model	30%	Week 08	Simulation Files & Report
Sustainable Basin Water Management Plan	35%	Week 12	Regional Policy & Engineering Plan
Environmental Compliance Field Log	10%	Continuous	Water Quality Sampling Logs

WEEKLY TERM SCHEDULE & READING REQUIREMENTS

Timeline	Technical Topic Focus	Assigned Readings & Docs
Week 01	Module 01: Hydrological Cycle & Precipitation	Chow, Maidment & Mays (Ch. 1–3)
Week 02	Module 02: Evapotranspiration & Soil Moisture	Penman (1948); Monteith (1965)
Week 03	Module 03: Infiltration & Green-Ampt Models	Chow (Ch. 4)
Week 04	Module 04: Surface Runoff & Unit Hydrographs	Sherman (1932); USGS Water Papers
Week 05	Module 05: Open Channel Flow & Manning Equation	Henderson (Ch. 2–4)
Week 06	Module 06: River Routing: Muskingum-Cunge	Cunge (1969)
Week 07	Module 07: Groundwater Hydrology & Darcy Law	Freeze & Cherry (Ch. 2 & 3)
Week 08	Module 08: Watershed Modeling with HEC-HMS	HEC-HMS User Manual & Specs
Week 09	Module 09: Urban Stormwater Management	EPA Stormwater Management Guidelines
Week 10	Module 10: Water Quality & Dissolved Oxygen Sag	Streeter & Phelps (1925)
Week 11	Module 11: Climate Change Impacts on River Basins	Milly et al. (2008)
Week 12	Module 12: Final Watershed Defense & Review	Hydrological Basin Report